

Northshore University Hospital

Department of Pathology, Neuropathology Division
880 Lakeside Drive, Riverwood, IL 60001 • CLIA 14D9911223

NEUROPATHOLOGY — INTRA-OPERATIVE AND PERMANENT SECTIONS

Patient: MARTINEZ-LANG, ELENA J.	Accession: SP-26-N-04412
MRN: MRN-3318477	Collected: 2026-04-04
DOB / Age / Sex: 1979-06-02 / 47 y / Female	Reported: 2026-04-08
Ordering provider: Sokolov, Dimitri R., MD (Neurosurgery)	Location: Inpatient — Neuroscience Step-Down

SPECIMEN

Right frontal lobe mass, gross total resection. Multiple fragments of soft gray-tan tissue, aggregate 4.1 x 3.2 x 2.6 cm, with focal hemorrhage. Submitted entirely (cassettes A1-A12). Material was also sent for molecular profiling and methylation classifier under separate accessions.

CLINICAL HISTORY

47-year-old woman with six-week history of new-onset focal motor seizures and progressive right-hand clumsiness. MRI brain demonstrated a 3.8 cm right frontal lobe mass with patchy enhancement and moderate vasogenic edema. No prior cancer history, no prior CNS irradiation. Family history negative for CNS tumors. Underwent right frontal craniotomy with gross total resection on 2026-04-04.

INTRA-OPERATIVE CONSULTATION

Smear preparations from the frontal lobe mass demonstrate an infiltrating glial neoplasm with mild-to-moderate atypia. Frozen section: "glial neoplasm; defer to permanent sections and ancillary studies for grading and classification." Conveyed verbally to Dr. Sokolov, 2026-04-04 09:46.

MICROSCOPIC DESCRIPTION

Sections demonstrate a moderately-to-markedly hypercellular astrocytic proliferation infiltrating cerebral cortex and subcortical white matter in a diffuse pattern. The tumor cells exhibit moderate nuclear pleomorphism with elongated, irregular nuclei and visible nucleoli; cytoplasm is scant to moderate. Mitotic figures are readily identified at up to 4 per 10 high-power fields in the most cellular areas (representative slide A4). Microvascular proliferation is NOT identified after review of multiple slides. Tumor necrosis is NOT identified. No oligodendroglial morphology is appreciated and there is no perinuclear halo formation. The infiltration pattern is consistent with a diffuse glioma rather than a circumscribed entity. Reactive astrocytosis and scattered macrophages are present at the tumor-brain interface.

IMMUNOHISTOCHEMISTRY

Marker	Result	Comment
GFAP	Diffuse, strong positivity	Confirms glial origin
IDH1 R132H (mAb H09)	Strong cytoplasmic positivity	Mutation-specific; correlated with NGS (separate accession)
ATRX	Loss of nuclear expression in tumor cells	Endothelial cells serve as internal positive control
p53	Strong nuclear positivity in >75% of tumor cells	Mutant pattern; correlation with NGS for variant call
Ki-67 (MIB-1)	~15% in hotspots, ~5% overall	Elevated for astrocytic glioma
Olig2	Positive	Glial lineage marker
EMA	Negative	Excludes meningioma / ependymoma
Synaptophysin	Negative in tumor cells	Excludes neuronal differentiation
BRAF V600E (VE1)	Negative	Performed given differential

FLUORESCENCE IN SITU HYBRIDIZATION (FISH) — 1p / 19q

Dual-color interphase FISH performed on FFPE sections (cassette A6) using paired centromeric and locus-specific probes for 1p36/1q25 and 19q13/19p13. 100 tumor nuclei scored.

Probe pair	Ratio (target/control)	Cutoff	Result
1p36 / 1q25	0.96	<0.80 = deletion	INTACT (no deletion)

19q13 / 19p13	0.94	<0.80 = deletion	INTACT (no deletion)
---------------	------	------------------	-----------------------------

INTERPRETATION: 1p/19q non-codeleted (intact). This finding excludes oligodendroglioma (which by current criteria requires both an IDH mutation and a whole-arm 1p/19q codeletion). Combined with the IDH1 R132H immunopositivity and ATRX loss, the case is in the diffuse astrocytic line.

DIAGNOSIS

RIGHT FRONTAL LOBE, MASS, GROSS TOTAL RESECTION:

- **DIFFUSE ASTROCYTIC GLIOMA, IDH-MUTANT (BY IHC AND PENDING NGS).**
- **Mitotic activity present (up to 4/10 HPF); microvascular proliferation and necrosis NOT identified.**
- **ATRX loss, p53 mutant pattern; 1p/19q intact.**
- **Tentative grading consistent with CNS WHO grade 3; final grading per integrated diagnosis once NGS and methylation classifier results return.**

COMMENT

Morphology, immunoprofile, and 1p/19q FISH support an IDH-mutant diffuse astrocytoma. Under WHO CNS5 (2021), grading of IDH-mutant astrocytomas integrates histology, molecular features (notably CDKN2A/B homozygous deletion as a grade-4 criterion), and occasionally methylation profiling. The histologic features (mitoses present, no MVP, no necrosis) place this at the upper end of grade 2-3 territory. Final classification is pending NGS (accession HX-NGS-CNS-018244) and methylation classifier (accession BTMRS-26-M-0907). An integrated layered diagnosis will follow.

Electronically signed: Reyes, Marisol G., MD, PhD — Neuropathology. Resident review: Iyer, P., MD (PGY-3). Report status: FINAL (surgical pathology component). This component report is part of an integrated diagnostic episode.

Page 1 of 1 • Accession SP-26-N-04412 • MRN-3318477